Anna-Drishti — Verification Sheet

Every number the simulation uses, in plain language. No code.

Smart India Hackathon 2026 · Internal Final · VIT — prepared for review, 22 August

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HOW TO USE THIS

Each item states a value the simulation currently uses and what it is claimed to mean. Tick **CORRECT** if you are willing to defend that number to a judge. Tick **CHANGE TO** and write the right value if it is wrong. Add the source you are working from — on Monday you will be asked where each number came from, and "it was in the code" is not an answer.

If you are unsure about an item, say so rather than ticking Correct. An unverified number we know about is manageable; one we think is verified is not.

Hand the pages back to Chiranjib and he will update the model.

DEADLINE

Return your pages by **SUNDAY 12:00**. The build freezes at 14:00 Sunday and no number changes after that — anything unverified by then gets removed from what we claim on stage.

Krittika Parida — Biology & realism

You own every biological number in the model. Fourteen items.

1 · FUNGAL GROWTH THRESHOLDS (ASPERGILLUS FLAVUS)

1.1 Minimum water activity for growth. Below this the model allows no fungal growth at all. `a_w = 0.82` *Is 0.82 the right floor, or should it be 0.85?*

☐ Correct ☐ Change to ....

Source: ....

1.2 Cardinal temperatures for growth — below the minimum or above the maximum, no growth. `min 12 °C · optimum 33 °C · max 43 °C`

☐ Correct ☐ Change to ....

Source: ....

1.3 Fastest possible growth, at perfect conditions, gives visible mould in `5 days` *Too fast, too slow, or right for grain (not agar)?*

☐ Correct ☐ Change to ....

Source: ....

2 · WHAT THE GROWTH MODEL PREDICTS — THE NUMBERS WE SAY OUT LOUD

These come straight out of items 1.1–1.3. If the table below looks wrong, the cardinal values above are wrong. **This table is the single most important thing on your pages** — a judge will ask where "mould in N days" came from.

WATER ACTIVITY	TEMPERATURE	MODEL SAYS MOULD ONSET IN	CORRECT?
0.99	33 °C	5.0 days	
0.95	30 °C	6.7 days	
0.92	30 °C	8.7 days	
0.90	30 °C	10.8 days	
0.87	30 °C	17.4 days	
0.85	30 °C	28.9 days	
0.83	30 °C	86.8 days	
0.87	20 °C	27.6 days	

☐ Whole table is defensible ☐ Corrections noted above

Source for safe-storage times: ....

3 · AFLATOXIN PRODUCTION

3.1 Toxin production needs a higher water activity than growth alone. `a_w ≥ 0.85`

☐ Correct ☐ Change to ....

Source: ....

3.2 Temperature band in which toxin production is possible. `25 °C to 35 °C` *I widened this from 30 to 35. The proposal cites 25–30 °C, but that is the reported optimum, not the limit — with 30 as a hard ceiling, a 30 °C godown that warmed even slightly fell out of the band and the model never predicted toxin risk at all. Is 35 °C a fair upper limit for production?*

☐ Correct ☐ Change to ....

Source: ....

4 · MOISTURE, WATER ACTIVITY AND THE SORPTION ISOTHERM

4.1 We convert grain moisture to water activity with the **modified Chung–Pfost** equation, using published shelled-corn coefficients (ASAE D245). *Right model and right commodity for what we claim?*

☐ Correct ☐ Change to ....

Source: ....

4.2 What that equation produces. Check these against your sorption tables:

GRAIN MOISTURE (WET BASIS)	TEMPERATURE	MODEL GIVES A_W	CORRECT?
12 %	30 °C	0.657	
14 %	25 °C	0.774	
16 %	30 °C	0.875	
18 %	25 °C	0.926	
20 %	30 °C	0.963	

☐ All defensible ☐ Corrections noted

4.3 Starting moisture of the sound grain in the stack. `12.0 % wet basis` ***I changed this from 13.5 % and it needs your sign-off.*** *At 13.5 % the healthy grain respires hard enough to mask the spoilage pocket entirely — and because the model is linear in respiration, no amount of adjusting the respiration rate fixes it. Only drier bulk grain does. 12 % is FCI wheat acceptance moisture and gives a_w 0.66, well below the growth floor. Is calling this "safely stored grain" defensible?*

☐ Correct ☐ Change to ....

Source: ....

5 · RESPIRATION

5.1 Baseline respiration of sound grain at 14 % moisture, 25 °C. `0.5 mg CO₂ / kg dry matter / day` *Published range is roughly 0.1–1. Is 0.5 sensible?*

☐ Correct ☐ Change to ....

Source: ....

5.2 Temperature sensitivity of respiration. $Q_{10} = 2.5$ *Rate multiplies by 2.5 for every 10 °C rise.*

☐ Correct ☐ Change to ....

Source: ....

5.3 Moisture sensitivity — respiration roughly **doubles per 1 % moisture** gained. Resulting rates, all at 30 °C:

GRAIN MOISTURE	RESPIRATION (MG CO_2 / KG / DAY)	CORRECT?
12 %	0.2	
16 %	3.2	
18 %	13	
20 %	53	

☐ All defensible ☐ Corrections noted

Source: ....

6 · DRY MATTER LOSS

6.1 Loss thresholds we report against. `0.5 % general · 1 % food grade · 0.04 % premium`

☐ Correct ☐ Change to ....

Source: ....

6.2 Conversion from CO_2 produced to dry matter lost. `1 % loss = 14.67 g CO₂ per kg` *This is fixed respiration stoichiometry, not a tuning choice — 180 g glucose yields 264 g CO_2. Confirm the chemistry only.*

☐ Correct ☐ Change to ....

Signed	Date	Items I could **not** verify
....	....	....

Netra R — Inspection baseline & evidence

You own the comparison arm. This is the most likely thing a judge attacks, and it lands on you. Eight items.

WHY THIS MATTERS MORE THAN THE OTHER SHEETS
Our headline claim is "conventional inspection found it on day 42, we flagged it on day 4." That claim is only as strong as the model of conventional practice underneath it. If it is pessimistic, the whole comparison collapses under one question — and a rigged comparison does more damage than no comparison at all. Push back hard on anything below that flatters us.

1 · HOW OFTEN, AND HOW DEEP

1.1 Inspection interval for a bagged stack. `every 14 days` *Is fortnightly right for an Indian godown? What does FCI actually specify, and what happens in practice?*

☐ Correct ☐ Change to ....

Source: ....

1.2 How far into a stack a spear sampler reaches. `0.4 m from any face` *This single number decides the whole comparison. Our demo pocket sits 1.6 m in, so it is physically unreachable. Is 0.4 m the honest reach of a bag spear?*

☐ Correct ☐ Change to ....

Source: ....

1.3 If a pocket *is* within reach, the chance an inspection actually finds it. `60 % per inspection` *Accounts for insects and spoilage being non-uniformly distributed. Too harsh, too generous?*

☐ Correct ☐ Change to ....

Source: ....

2 · WHAT COUNTS AS DETECTION

2.1 Mould visible on an outer face is **always** caught at the next inspection. *Is a certainty fair, or should this have a miss rate too?*

☐ Correct ☐ Change to ....

2.2 Second detection route, which **I added and which needs your judgement.** Once a colony is well established, the stack smells musty and bags feel warm, so it is noticed at the next routine inspection even if no spear can reach it. *Without this route, a deep pocket read "never detected in 60 days." That is arguably true to FAO — but it looks rigged. With it, conventional detection lands at day 42. Is that route real, and is the timing fair?*

☐ Fair as is ☐ Change to ....

Source: ....

3 · THE WRITTEN JUSTIFICATION — NOT YET STARTED

The model is coded. The written defence of it does not exist and only you can write it. It needs to answer, in prose, with citations:

- How often inspection happens and on whose authority
- Which part of a stack a spear can physically reach, and which it cannot

- Why absence of live adults in a sample does not mean absence of infestation
- Why insects concentrate in pockets rather than distributing evenly

☐ Drafted ☐ Sent to Chiranjib

4 · ENTOMOLOGY FIGURES CARRIED FROM THE PROPOSAL

4.1 One insect per 100 kg can develop into a damaging population in roughly four months under humid tropical conditions. *Still stands? We quote it on stage.*

☐ Correct ☐ Change to ....

Source: ....

4.2 The question you will be asked on Monday, almost certainly in these words: **"Isn't your baseline unfairly pessimistic?"** *Write your answer here. It should cite a real inspection interval and a real detection threshold, not an opinion.*

....

....

....

Signed	Date	Items I could **not** verify
....	....	....

Devavrath Ramesh — Scale, transport & deployment

You own the physical numbers: how big the stack is, how fast things move through it, and where the probes go. Twelve items.

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1 · SCALE CALIBRATION

1.1 Stack dimensions the simulation represents. `4.0 m wide × 2.5 m tall` *Does that match a real godown bag stack?*

☐ Correct ☐ Change to ....

1.2 Resolution of the model grid. `1 cell = 10 cm · 1 step = 0.25 hours`

☐ Correct ☐ Change to ....

2 · TRANSPORT COEFFICIENTS — THE CORE OF OUR ARGUMENT

Our entire thesis is that CO_2 outruns heat, which outruns moisture. **The ratios matter far more than the absolute values.**

QUANTITY	VALUE (M^2/S)	TIME TO CROSS 0.4 M	CORRECT?
CO_2 in interstitial air	2.0×10^{-6}	0.9 days	
Heat through bulk grain	1.2×10^{-7}	15.4 days	
Moisture migration	1.5×10^{-8}	123.5 days	

☐ All three defensible ☐ Corrections noted

Sources: ....

QUESTION YOU WILL BE ASKED

“Is your moisture figure a molecular diffusivity?” — No. It is an **EFFECTIVE** coefficient lumping convective moisture migration together with diffusion, because in a real bag stack convection dominates. Molecular diffusion alone would be far slower. Make sure you can say this in your own words.

☐ I can defend this

3 · BULK GRAIN PROPERTIES

3.1 Bulk density of bagged grain. `800 kg/m³`

☐ Correct ☐ Change to ....

3.2 Porosity — the fraction of the bed that is air, and therefore where CO_2 accumulates. `0.40`

☐ Correct ☐ Change to ....

3.3 Specific heat capacity, which sets how much a hotspot warms. `1800 J / kg·K`

☐ Correct ☐ Change to ....

4 · ENVIRONMENT

4.1 Ambient godown temperature. `30 °C` *Note: the model holds this steady, with no day/night cycle. Is that acceptable?*

☐ Correct ☐ Change to ....

4.2 Ambient CO_2 outside the stack. `400 ppm`

☐ Correct ☐ Change to ....

4.3 How fast the exposed faces of the stack vent CO_2 to the surrounding air. `time constant ≈ 0.5 days` *This sets the background CO_2 level inside the stack — currently around 1,800 ppm for sound grain. Plausible for a bag stack in an open godown?*

☐ Correct ☐ Change to ....

Source: ....

5 · PROBE GEOMETRY — YOURS TO JUSTIFY

5.1 Lance positions across a 4 m stack face. `0.8 m · 2.0 m · 3.2 m` *I picked these to space three lances evenly. Why this spacing, and is three the right number?*

☐ Correct ☐ Change to ....

5.2 Sensor depths on each lance. `0.5 m · 1.2 m · 2.0 m below the top` *Note 1.2 m is the deepest point from any face and reads highest CO_2 naturally — the detector accounts for that. Are these the right depths?*

☐ Correct ☐ Change to ....

6 · COVERAGE ANALYSIS — NOT YET STARTED

This is your genuine engineering finding and nobody else can produce it. Method: open the demo, inject a hotspot at many different positions, record how many days each takes to be flagged. Produce one sentence of the form:

> *"Three lances at this spacing detect N % of hotspots within X days."*

☐ Data collected ☐ Chart made Result: ....

Signed	Date	Items I could **not** verify
....	....	....

Mohit Kumar & Vishal Vivek — Physics engine and detector

You two can read the code directly, so these are the claims you will be asked to defend rather than numbers to look up.
Repo: github.com/Chiranjib-x/anna-drishti

MOHIT — THE PHYSICS MODEL

M1 The solver is an explicit finite-difference scheme with zero-flux boundaries. It is only stable while `D·dt/dx² ≤ 0.25` — currently 0.18. *Understand why this limit exists and what happens if it is exceeded. If anyone ever sees NaN on screen, this is the first thing to check.*

☐ I can defend this

M2 A single respiration term drives **both** CO_2 production and heat release — they are not independent effects, they are the same biology seen two ways. *This is the core coupling. Be ready to explain it.*

☐ I can defend this

M3 Heat released per unit of CO_2 produced. `10,670 J per g CO₂` Respiration is capped at 4,000 mg/kg/day to stop runaway self-heating. *Is the cap ever reached in a normal run? Check.*

☐ Checked ☐ Issue found ....

VISHAL — SENSING AND THE DETECTOR

V1 Sensor noise applied to every reading, so the detector runs on imperfect data: `CO₂ ±25 ppm +1.5 % · RH ±1.0 % · temp ±0.15 °C` *Do these match real NDIR and digital RH sensor datasheets? Check against the parts in the ₹20,000 budget.*

☐ Correct ☐ Change to ....

V2 CO_2 is compared **across lances at the same depth**, not against all nine ports. *Depth stratification is real and about 1,000 ppm — comparing all nine ports buried the signal completely. This is the change most likely to be questioned.*

☐ I can defend this

V3 Alert fires at `3.0 σ, sustained over 3 consecutive readings` *Measured separation: clean grain peaks at 1.5 σ, a real hotspot reaches 4.2 σ. Verify by running the demo on clean grain for 60 days and confirming it stays silent.*

☐ Verified silent on clean grain ☐ False alarm seen ....

V4 Risk score weighting. `35 % water activity · 40 % urgency · 25 % CO₂ anomaly` *Alert level is 45 out of 100. Sensible weighting?*

☐ Correct ☐ Change to ....

Mohit — signed .... Vishal — signed .... Date ....